

Azure AI Search Deep Dive (Solution Architect Track)

Why Azure AI Search?

Imagine:

100,000 Project Documents

50,000 PDFs

10,000 Runbooks

5,000 Support Guides

GPT alone cannot search all these effectively.

Azure AI Search helps:

Store

Index

Search

Retrieve

Rank

Filter

enterprise content.

Part 1 - Theory (60 mins)

Topic 1 - What is Azure AI Search?

Azure AI Search is Microsoft's enterprise search platform.

It enables:

- Keyword Search
- Semantic Search
- Vector Search
- Hybrid Search

Architecture

```
graph TD; Documents --> AzureAI[Azure AI Search]; AzureAI --> Results;
```

Documents

↓

Azure AI Search

↓

Results

Enterprise Example

- HR Documents
- Project Documents
- Support Guides
- Architecture Documents

All searchable from a single index.

Topic 2 - Core Components

Azure AI Search consists of:



Topic 3 - Data Sources

Data Source = Where data comes from.

Examples:

Azure Blob Storage
SQL Database
Cosmos DB
SharePoint
ADLS

Example

Project PDFs
Stored In
Blob Storage

Topic 4 - Index

The Index is the heart of Azure AI Search.

Think:

Searchable Database
for retrieval.

Example

Document:

AzureOpenAIGuide.pdf

Indexed as:

Title
Content
Author
Created Date
Project

Topic 5 - Fields

Fields define searchable content.

Example:

DocumentId
Title
Content
ProjectName
CreatedDate

Searchable Fields

Examples:

Title

Content

Filterable Fields

Examples:

Department

Project

Author

Topic 6 - Indexers

Indexers automate ingestion.

Flow

Blob Storage

↓

Indexer

↓

Azure AI Search Index

Benefits

Automatic:

✓ Crawling

✓ Indexing

✓ Updating

Topic 7 - Skillsets

Skillsets enrich data.

Examples:

OCR

Language Detection

Translation

Entity Recognition

Example

PDF Image

Image

↓

OCR Skill

↓

Extracted Text

↓

Index

Topic 8 - Semantic Search

Keyword Search:

Azure OpenAI

finds exact terms.

Semantic Search:

How to create OpenAI service?

matches:

Azure OpenAI Setup Guide

based on meaning.

Topic 9 - Vector Search

Learned yesterday.

Azure AI Search also supports:

Embeddings

Vector Storage

Vector Retrieval

Topic 10 - Hybrid Search

Enterprise best practice.

Hybrid Search:

Keyword Search

+

Vector Search

Architecture

Query

↓

Keyword Search

↓

Vector Search

↓

Merged Results

Why Hybrid Search?

Highest accuracy.

Topic 11 - AI Search in RAG

Tomorrow's topic.

But understand this first.

RAG Architecture

User Question

↓

Embedding

↓

AI Search

↓

Relevant Chunks

↓

Azure OpenAI

↓

Answer

Topic 12 - Enterprise Search Architecture

Example:

HR Assistant

Documents:

Policies

Leave Rules

Benefits

Travel Policies

Architecture

Documents



Blob Storage



Indexer



AI Search



Retriever



GPT



Response